Chapter 1

Introduction

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 1; printed pp. 1–13; PDF pages 18–30.

Chinese reference: 中译本：引言，书内第 1–9 页；PDF 第 19–27 页。

Editorial convention. The source exposition and all five source problems are retained below. Chinese margin notes and passages headed "Editorial supplement" are added by the editor. Only the two equations numbered (1) and (2) in the source carry those source numbers here; supplementary displays are unnumbered.

This chapter surveys the principal theoretical issues concerning the solving of partial differential equations.

To follow the subsequent discussion, the reader should first of all turn to Appendix A and look over the notation presented there, particularly the multiindex notation for partial derivatives.

1.1 Partial Differential Equations

A *partial differential equation* (PDE) is an equation involving an unknown function of two or more variables and certain of its partial derivatives.

Using the notation explained in Appendix A, we can write out symbolically a typical PDE, as follows. Fix an integer $k \geq 1$ and let U denote an open subset of $\mathbb{R}^n$.

Definition (source, §1.1). An expression of the form

$$F(D^k u(x), D^{k-1}u(x), \dots, Du(x), u(x), x) = 0 \quad (x \in U) \tag{1}$$

is called a *kth-order partial differential equation*, where

$$F : \mathbb{R}^{n^k} \times \mathbb{R}^{n^{k-1}} \times \cdots \times \mathbb{R}^n \times \mathbb{R} \times U \to \mathbb{R}$$

is given and $u : U \to \mathbb{R}$ is the unknown.

We solve the PDE if we find all u verifying (1), possibly only among those functions satisfying certain auxiliary boundary conditions on some part Γ of ∂U. By finding the solutions we mean, ideally, obtaining simple, explicit solutions, or, failing that, deducing the existence and other properties of solutions.

Definitions (source, §1.1).

(i) The partial differential equation (1) is called *linear* if it has the form

$$\sum_{|\alpha| \le k} a_\alpha(x) D^\alpha u = f(x)$$

for given functions a_α $(|\alpha| \le k), f$. This linear PDE is *homogeneous* if $f \equiv 0$.

(ii) The PDE (1) is *semilinear* if it has the form

$$\sum_{|\alpha| = k} a_\alpha(x) D^\alpha u + a_0(D^{k-1}u, \dots, Du, u, x) = 0.$$

(iii) The PDE (1) is *quasilinear* if it has the form

$$\sum_{|\alpha| = k} a_\alpha(D^{k-1}u, \dots, Du, u, x) D^\alpha u + a_0(D^{k-1}u, \dots, Du, u, x) = 0.$$

(iv) The PDE (1) is *fully nonlinear* if it depends nonlinearly upon the highest order derivatives.

导数数组与对称性

原书把 $D^k u$ 写成含 n^k 个分量的数组。若 $u \in C^k$，混合偏导可交换，独立分量只有 $\binom{n+k-1}{k}$ 个。前者按求导次序储存，后者按多重指标计数；两种记号并不矛盾。

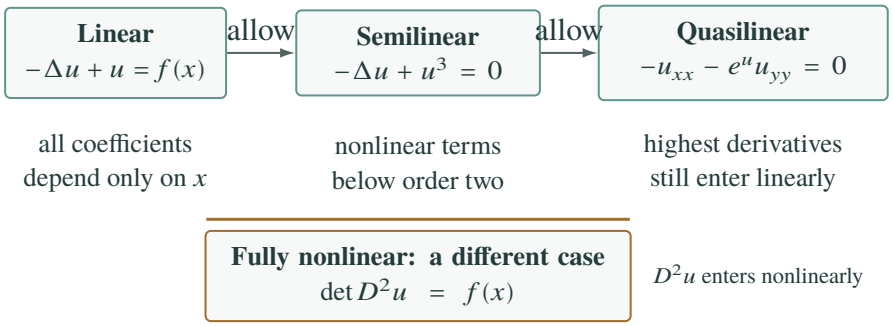


Diagram T1.1. Classify by the highest derivatives. The first three classes allow successively more dependence on lower-order variables. In the last example the highest derivatives themselves enter nonlinearly. All displayed examples are second order.

分类看最高阶导数

先找阶数 k，再看 $D^k u$ 如何出现。半线性：最高阶系数只依赖 x；拟线性：系数可依赖低阶导数。例如 $u_t + uu_x = 0$ 是一阶拟线性，而 $u_t + uu_x + u_{xxx} = 0$ 是三阶半线性。

A *system of partial differential equations* is, informally speaking, a collection of several PDE for several unknown functions.

Definition (source, §1.1). Let k be a positive integer. An expression of the form

$$\mathbf{F}(D^k\mathbf{u}(x), \dots, D\mathbf{u}(x), \mathbf{u}(x), x) = \mathbf{0} \quad (x \in U) \tag{2}$$

is called a *kth-order system of partial differential equations*, where

$$\mathbf{F} : \mathbb{R}^{mn^k} \times \mathbb{R}^{mn^{k-1}} \times \cdots \times \mathbb{R}^{mn} \times \mathbb{R}^m \times U \to \mathbb{R}^m$$

is given and

$$\mathbf{u} : U \to \mathbb{R}^m, \qquad \mathbf{u} = (u^1, \dots, u^m)$$

is the unknown.

Here we are supposing that the system comprises the same number m of scalar equations as unknowns $(u^1, \dots, u^m)$. This is the most common circumstance, although other systems may have fewer or more equations than unknowns. Systems are classified in the obvious way as being linear, semilinear, etc.

Notation. We write "PDE" as an abbreviation for both the singular "partial differential equation" and the plural "partial differential equations".

Editorial supplement: the multiindex convention

For $\alpha = (\alpha_1, \dots, \alpha_n) \in \{0, 1, 2, \dots\}^n$, set

$$|\alpha| = \sum_{i=1}^{n} \alpha_i, \qquad D^\alpha u = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_n}^{\alpha_n} u.$$

The *principal part* of a linear operator of order k is $\sum_{|\alpha|=k} a_\alpha(x) D^\alpha$; the terms of smaller order are its lower-order terms. Here $D^0 u = u$. For example, when $n = 2$, $D^{(2,1)}u = u_{x_1 x_1 x_2}$ and $|(2,1)| = 3$.

The words linear, semilinear and quasilinear describe nested forms. When a single classification is requested, we report the most restrictive applicable form: linear first, then semilinear, then quasilinear. These definitions concern the displayed differential operator. An algebraic rewriting that changes its form need not preserve its classification.

等价零点不保结构

把 $Lu = 0$ 改写成 $(Lu)^3 = 0$，经典解集不变，但新表达式对最高阶导数非线性，且在线性化时可能退化。因此分类必须说明采用哪个算子表达式，不能只看方程的解集。

1.2 Examples

There is no general theory known concerning the solvability of all partial differential equations. Such a theory is extremely unlikely to exist, given the rich variety of physical, geometric, and probabilistic phenomena which can be modeled by PDE. Instead, research focuses on various particular partial differential equations that are important for applications within and outside of mathematics, with the hope that insight from the origins of these PDE can give clues as to their solutions.

Following is a list of many specific partial differential equations of interest in current research. This listing is intended merely to familiarize the reader with the names and forms of various famous PDE. To display most clearly the mathematical structure of these equations, we have mostly set relevant physical constants to unity. We will later discuss the origin and interpretation of many of these PDE.

Throughout $x \in U$, where U is an open subset of $\mathbb{R}^n$, and $t \geq 0$. Also $Du = D_x u = (u_{x_1}, \dots, u_{x_n})$ denotes the gradient of u with respect to the spatial variable $x = (x_1, \dots, x_n)$. The variable t always denotes time.

1.2.1 Single partial differential equations

a. Linear equations

1. Laplace's equation
$$\Delta u = \sum_{i=1}^{n} u_{x_i x_i} = 0.$$
2. Helmholtz's (or eigenvalue) equation
$$-\Delta u = \lambda u.$$
3. Linear transport equation
$$u_t + \sum_{i=1}^{n} b^i u_{x_i} = 0.$$
4. Liouville's equation
$$u_t + \sum_{i=1}^{n} (b^i u)_{x_i} = 0.$$

5. Heat (or diffusion) equation

$$u_t - \Delta u = 0.$$

6. Schrödinger's equation

$$\mathrm{i}u_t + \Delta u = 0.$$

7. Kolmogorov's equation

$$u_t - \sum_{i,j=1}^{n} a^{ij} u_{x_i x_j} + \sum_{i=1}^{n} b^i u_{x_i} = 0.$$

8. Fokker–Planck equation

$$u_t - \sum_{i,j=1}^{n} (a^{ij} u)_{x_i x_j} - \sum_{i=1}^{n} (b^i u)_{x_i} = 0.$$

9. Wave equation

$$u_{tt} - \Delta u = 0.$$

10. Klein–Gordon equation

$$u_{tt} - \Delta u + m^2 u = 0.$$

11. Telegraph equation

$$u_{tt} + 2du_t - u_{xx} = 0.$$

12. General wave equation

$$u_{tt} - \sum_{i,j=1}^{n} a^{ij} u_{x_i x_j} + \sum_{i=1}^{n} b^i u_{x_i} = 0.$$

13. Airy's equation

$$u_t + u_{xxx} = 0.$$

14. Beam equation

$$u_{tt} + u_{xxxx} = 0.$$

b. Nonlinear equations

1. Eikonal equation

$$|Du| = 1.$$

2. Nonlinear Poisson equation

$$-\Delta u = f(u).$$

随流不变与质量守恒

输运方程沿 $\dot{X} = b(X,t)$ 保持 $u(X(t),t)$ 不变。Liouville 方程展开后多出 $(\operatorname{div} b)u$，描述密度随体积伸缩而变。只有 $\operatorname{div} b = 0$ 时两式一致。

算子与形式伴随

令 $L = a^{ij}\partial_{ij} - b^i\partial_i$（重复指标求和）。分部积分给出 $L^*\rho = \partial_{ij}(a^{ij}\rho) + \partial_i(b^i\rho)$。两式分别为 $u_t = Lu$ 与 $\rho_t = L^*\rho$，漂移项符号因此不同。

阶数与时间初值

热方程、波方程都是二阶 PDE，但前者时间一阶，后者时间二阶，通常分别需给 $u(0)$ 与 $(u(0), u_t(0))$。梁方程总阶数为四、时间阶数为二；总阶数不能直接决定初值个数。

3. p-Laplacian equation

$$\operatorname{div}(|Du|^{p-2}Du) = 0.$$

4. Minimal surface equation

$$\operatorname{div}\left(\frac{Du}{(1+|Du|^2)^{1/2}}\right) = 0.$$

5. Monge–Ampère equation

$$\det(D^2u) = f.$$

6. Hamilton–Jacobi equation

$$u_t + H(Du, x) = 0.$$

7. Scalar conservation law

$$u_t + \operatorname{div}\mathbf{F}(u) = 0.$$

8. Inviscid Burgers' equation

$$u_t + uu_x = 0.$$

9. Scalar reaction-diffusion equation

$$u_t - \Delta u = f(u).$$

10. Porous medium equation

$$u_t - \Delta(u^\gamma) = 0.$$

11. Nonlinear wave equation

$$u_{tt} - \Delta u + f(u) = 0.$$

12. Korteweg–deVries (KdV) equation

$$u_t + uu_x + u_{xxx} = 0.$$

13. Nonlinear Schrödinger equation

$$\mathrm{i}u_t + \Delta u = f(|u|^2)u.$$

退化与奇异

对 p-Laplacian，$p > 2$ 时主部在 $Du = 0$ 退化；$1 < p < 2$ 时其线性化系数在那里奇异。散度形式及相应弱形式仍有意义，不能把只在 $Du \neq 0$ 合法的展开直接用于临界点。

多孔介质的扩散系数

在 $u > 0$ 的光滑区，$\Delta(u^\gamma) = \gamma u^{\gamma-1}\Delta u + \gamma(\gamma-1)u^{\gamma-2}|Du|^2$。通常取 $\gamma > 1, u \geq 0$，主部在 $u = 0$ 退化；$0 < \gamma < 1$ 对应快扩散情形。

1.2.2 Systems of partial differential equations

a. Linear systems

1. Equilibrium equations of linear elasticity

$$\mu\Delta\mathbf{u} + (\lambda + \mu)D(\operatorname{div}\mathbf{u}) = \mathbf{0}.$$

2. Evolution equations of linear elasticity

$$\mathbf{u}_{tt} - \mu\Delta\mathbf{u} - (\lambda + \mu)D(\operatorname{div}\mathbf{u}) = \mathbf{0}.$$

3. Maxwell's equations

$$\begin{cases}\mathbf{E}_t = \operatorname{curl}\mathbf{B},\\ \mathbf{B}_t = -\operatorname{curl}\mathbf{E},\\ \operatorname{div}\mathbf{B} = \operatorname{div}\mathbf{E} = 0.\end{cases}$$

b. Nonlinear systems

1. System of conservation laws

$$\mathbf{u}_t + \operatorname{div}\mathbf{F}(\mathbf{u}) = \mathbf{0}.$$

2. Reaction-diffusion system

$$\mathbf{u}_t - \Delta\mathbf{u} = \mathbf{f}(\mathbf{u}).$$

3. Euler's equations for incompressible, inviscid flow

$$\begin{cases}\mathbf{u}_t + \mathbf{u}\cdot D\mathbf{u} = -Dp,\\ \operatorname{div}\mathbf{u} = 0.\end{cases}$$

4. Navier–Stokes equations for incompressible, viscous flow

$$\begin{cases}\mathbf{u}_t + \mathbf{u}\cdot D\mathbf{u} - \Delta\mathbf{u} = -Dp,\\ \operatorname{div}\mathbf{u} = 0.\end{cases}$$

约束的传播

真空 Maxwell 系统中，$\partial_t \operatorname{div}\mathbf{E} = 0$、$\partial_t \operatorname{div}\mathbf{B} = 0$，因为散度作用于旋度为零。散度条件是初值必须满足且由演化保持的约束，不能视作任意额外数据。

See Zwillinger [Zw] for a much more extensive listing of interesting PDE.

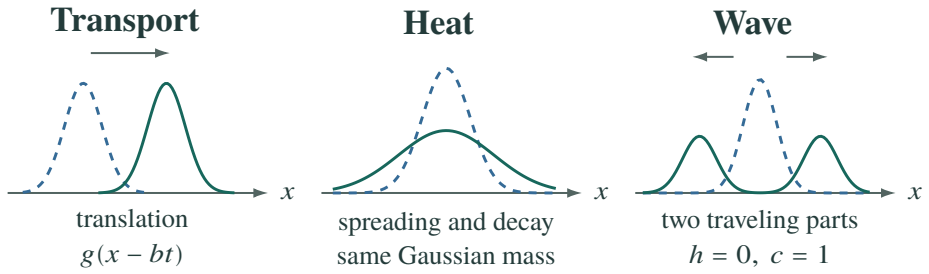


Diagram T1.2. Three evolution mechanisms. Dashed curves are initial profiles and solid curves are later profiles. Transport translates a shape; heat spreads a Gaussian while preserving its integral; the one-dimensional wave equation with zero initial velocity splits a profile into two half-amplitude translates. The panel axes are schematic and use separate scales.

Editorial supplement: two principal-part calculations

For $1 < p < \infty$ and a C^2 function at a point where $Du \neq 0$, the chain rule gives

$$\operatorname{div}(|Du|^{p-2}Du) = \sum_{i,j=1}^{n} A_{ij}(Du)u_{x_i x_j},$$

$$A(q) = |q|^{p-2}\left(I + (p-2)\frac{q \otimes q}{|q|^2}\right).$$

The eigenvalue in the direction of q is $(p-1)|q|^{p-2}$; in every orthogonal direction it is $|q|^{p-2}$. This proves ellipticity at $q \neq 0$ and explains its failure to be uniform as $|q| \to 0$ or $|q| \to \infty$ when $p \neq 2$. At $p = 2$ the equation is exactly Laplace's equation.

Similarly, the minimal surface equation expands as

$$\frac{1}{\sqrt{1 + |Du|^2}} \sum_{i,j=1}^{n} \left(\delta_{ij} - \frac{u_{x_i}u_{x_j}}{1 + |Du|^2}\right) u_{x_i x_j} = 0.$$

The matrix in parentheses has eigenvalues 1 transverse to Du and $(1 + |Du|^2)^{-1}$ parallel to Du. Thus it is positive definite for each finite gradient, and a bound on $|Du|$ supplies a uniform ellipticity bound. Both calculations exhibit quasilinear second-order equations.

压力是约束乘子

对光滑不可压流取散度，得到 $-\Delta p = \sum_{i,j}(\partial_i u^j)(\partial_j u^i)$。压力由速度及边界条件耦合确定；没有独立的压力演化方程。仅看速度的扩散项会漏掉这一非局部约束。

1.3 Strategies for Studying PDE

As explained in §1.1 our goal is the discovery of ways to solve partial differential equations of various sorts, but—as should now be clear in view of the many diverse examples set forth in §1.2—this is no easy task. And indeed the very question of what it means to "solve" a given PDE can be subtle, depending in large part on the particular structure of the problem at hand.

1.3.1 Well-posed problems, classical solutions

The informal notion of a well-posed problem captures many of the desirable features of what it means to solve a PDE. We say that a given problem for a partial differential equation is well-posed if

(i) the problem in fact has a solution;
(ii) this solution is unique; and
(iii) the solution depends continuously on the data given in the problem.

The last condition is particularly important for problems arising from physical applications: we would prefer that our (unique) solution changes only a little when the conditions specifying the problem change a little. (For many problems, on the other hand, uniqueness is not to be expected. In these cases the primary mathematical tasks are to classify and to characterize the solutions.)

Now clearly it would be desirable to "solve" PDE in such a way that (i)–(iii) hold. But notice that we still have not carefully defined what we mean by a "solution". Should we ask, for example, that a "solution" u must be real analytic or at least infinitely differentiable? This might be desirable, but perhaps we are asking too much. Maybe it would be wiser to require a solution of a PDE of order k to be at least k times continuously differentiable. Then at least all the derivatives which appear in the statement of the PDE will exist and be continuous, although maybe certain higher derivatives will not exist. Let us informally call a solution with this much smoothness a classical solution of the PDE: this is certainly the most obvious notion of solution.

So by solving a partial differential equation in the classical sense we mean if possible to write down a formula for a classical solution satisfying (i)–(iii) above, or at least to show such a solution exists, and to deduce various of its properties.

Monge–Ampère 的分支

当 $n \geq 2$，行列式非线性地依赖 Hessian。在 D^2u 正定的凸解分支上，线性化系数为正定的余子式矩阵。离开该分支，椭圆性没有保证；仅写 $\det D^2u = f$ 还不足以指定椭圆问题。

适定性依赖拓扑

连续依赖必须指定数据空间 X 和解空间 Y。常见目标是 $\|u-\tilde{u}\|_Y \leq C\|g-\tilde{g}\|_X$。同一方程更换范数、时间方向或边界条件后，适定性可能改变。

1.3.2 Weak solutions and regularity

But can we achieve this? The answer is that certain specific partial differential equations (e.g. Laplace's equation) can be solved in the classical sense, but many others, if not most others, cannot. Consider for instance the scalar conservation law

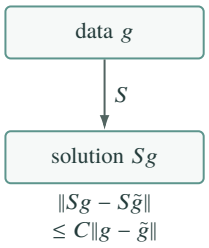
$$u_t + F(u)_x = 0.$$

We will see in §3.4 that this PDE governs various one-dimensional phenomena involving fluid dynamics, and in particular models the formation and propagation of shock waves. Now a shock wave is a curve of discontinuity of the solution u; and so if we wish to study conservation laws, and recover the underlying physics, we must surely allow for solutions u which are not continuously differentiable or even continuous. In general, as we shall see, the conservation law has no classical solutions but is well-posed if we allow for properly defined generalized or weak solutions.

This is all to say that we may be forced by the structure of the particular equation to abandon the search for smooth, classical solutions. We must instead, while still hoping to achieve the well-posedness conditions (i)–(iii), investigate a wider class of candidates for solutions. And in fact, even for those PDE which turn out to be classically solvable, it is often most expedient initially to search for some appropriate kind of weak solution.

The point is this: if from the outset we demand that our solutions be very regular, say k-times continuously differentiable, then we are usually going to have a really hard time finding them, as our proofs must then necessarily include possibly intricate demonstrations that the functions we are building are in fact smooth enough. A far more reasonable strategy is to consider as separate the existence and the smoothness (or regularity) problems. The idea is to define for a given PDE a reasonably wide notion of a weak solution, with the expectation that since we are not asking too much by way of smoothness of this weak solution, it may be easier to establish its existence, uniqueness, and continuous dependence on the given data. Thus, to repeat, it is often wise to aim at proving well-posedness in some appropriate class of weak or generalized solutions.

图 T1.M：连续依赖的方向



先说明数据与解各用什么范数，再检验解映射是否连续。存在、唯一和连续依赖是三个不同问题。

经典解可局部存在

“没有经典解”不能理解成任意初值、任意短时间都无解。光滑 Burgers 初值通常先产生局部经典解，再因特征线相交形成激波。失效的是一般情形下全局保持光滑的要求。

Now, as noted above, for various partial differential equations this is the best that can be done. For other equations we can hope that our weak solution may turn out after all to be smooth enough to qualify as a classical solution. This leads to the question of regularity of weak solutions. As we will see, it is often the case that the existence of weak solutions depends upon rather simple estimates plus ideas of functional analysis, whereas the regularity of the weak solutions, when true, usually rests upon many intricate calculus estimates.

弱形式与唯一性

分部积分把导数转移给光滑试验函数，降低对 u 的正则性要求。但守恒律的分布弱解通常不唯一；还需熵条件筛选物理解。“弱解存在”本身不等于问题已经适定。

Let me explicitly note here that once we are past Part I (Chapters 2–4), our efforts will be largely devoted to proving mathematically the existence of solutions to various sorts of partial differential equations, and not so much to deriving formulas for these solutions. This may seem wasted or misguided effort, but in fact mathematicians are like theologians: we regard existence as the prime attribute of what we study. But unlike theologians, we need not always rely upon faith alone.

1.3.3 Typical difficulties

Following are some vague but general principles, which may be useful to keep in mind:

(i) Nonlinear equations are more difficult than linear equations; and, indeed, the more the nonlinearity affects the higher derivatives, the more difficult the PDE is.
(ii) Higher-order PDE are more difficult than lower-order PDE.
(iii) Systems are harder than single equations.
(iv) Partial differential equations entailing many independent variables are harder than PDE entailing few independent variables.
(v) For most partial differential equations it is not possible to write out explicit formulas for solutions.

None of these assertions is without important exceptions.

Editorial supplement: estimates, instability, and shocks

The following three calculations make the preceding distinctions precise. They are editorial additions, not proofs stated in the source introduction.

1. A forward heat estimate. Let U be a bounded smooth domain, and let u, v be sufficiently smooth solutions of $u_t - \Delta u = 0$ with the same homogeneous Dirichlet boundary condition. For $w = u - v$, multiplication

by w and integration by parts give

$$\frac{1}{2}\frac{d}{dt}\int_U w^2\,dx + \int_U |Dw|^2\,dx = \int_{\partial U} w\,\partial_\nu w\,dS = 0.$$

Consequently, for every time within their common interval of existence,

$$\|u(t) - v(t)\|_{L^2(U)} \le \|u(0) - v(0)\|_{L^2(U)}.$$

This proves uniqueness in this smooth class and an L^2 continuous-dependence estimate. It does not prove existence. Extending the calculation to weak solutions requires a justification for the time derivative and the test function w; the formal identity alone is not that justification.

边界项决定估计

这里消失的是 $w\partial_\nu w$，依据为相同的齐次 Dirichlet 条件。若边值不同，就不能直接令边界项为零。同样的能量计算必须随边界条件调整，不能只从方程形式照抄结论。

2. Reversing time destroys this stability. On $U = (0, \pi)$ fix $T > 0$, with zero boundary values, and define

$$u_j(x,t) = e^{j^2(T-t)}\sin(jx), \qquad 0 \le t \le T.$$

Every u_j is a smooth forward heat solution. Its terminal and initial norms satisfy

$$\|u_j(T)\|_2 = \sqrt{\pi/2}, \qquad \|u_j(0)\|_2 = e^{j^2T}\sqrt{\pi/2}.$$

More decisively, $v_j = e^{-j^2T/2}u_j$ satisfies

$$\|v_j(T)\|_2 \longrightarrow 0, \qquad \|v_j(0)\|_2 \longrightarrow \infty.$$

Thus the map from terminal data to the initial solution cannot be continuous at zero in L^2, even when restricted to these smooth solvable data. No claim of nonexistence is needed for this instability argument.

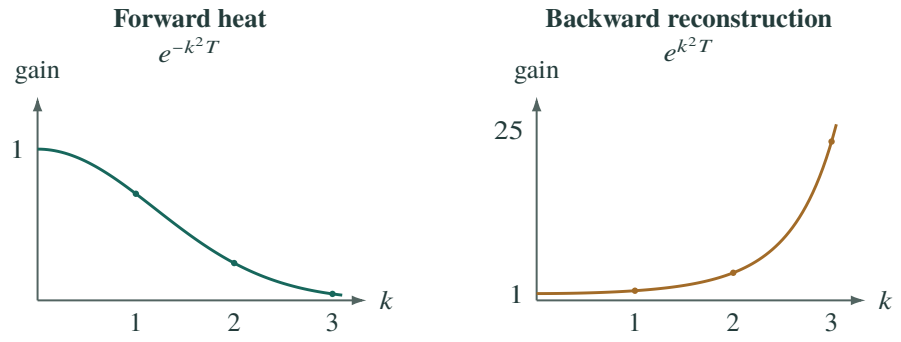


Diagram T1.3. High frequencies distinguish stability from solvability. For the Dirichlet mode $\sin(kx)$ on $(0, \pi)$, forward heat evolution multiplies amplitude by e^{-k^2T}; reconstruction multiplies by its reciprocal. Here $T = 0.35$ and vertical scales differ. Arbitrarily small terminal errors in high modes can therefore give large reconstructed initial errors.

3. Smooth data can create a Burgers singularity. For $u_t + uu_x = 0$ on the line, let $u(x,0) = g(x)$, where $g \in C^1(\mathbb{R})$ and both g and g' are bounded. Along a characteristic starting at ξ,

$$X(\xi,t) = \xi + tg(\xi), \qquad u(X(\xi,t),t) = g(\xi).$$

Before $\partial_\xi X = 1 + tg'(\xi)$ vanishes, inversion and the chain rule give

$$u_x(X(\xi,t),t) = \frac{g'(\xi)}{1+tg'(\xi)}.$$

If $g'(\xi_0) < 0$, this denominator vanishes at $t = -1/g'(\xi_0) > 0$, so the formula cannot continue as a C^1 solution through that event. If the negative minimum $m = \min_{\mathbb{R}} g'$ is attained, the first such time is $T_* = -1/m$. A bounded amplitude therefore does not prevent derivative blow-up.

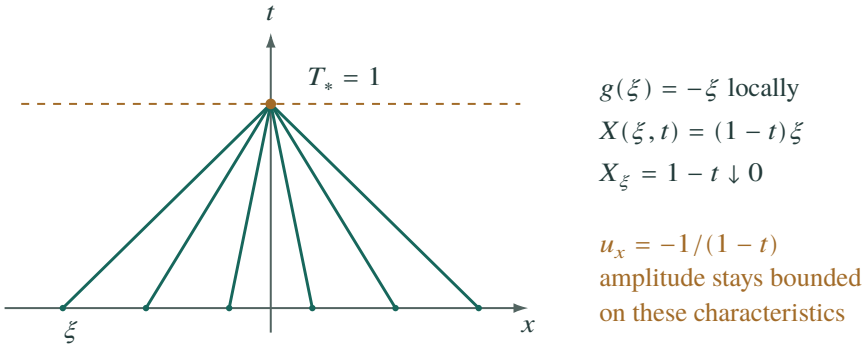


Diagram T1.4. Loss of invertibility precedes a shock. For the local initial profile $g(\xi) = -\xi$, the characteristics meet at $t = 1$. The characteristic formula is valid for $t < 1$; the meeting point depicts failure of a classical continuation, rather than several values of an admissible solution at one point.

To include a discontinuity, write $F(u) = u^2/2$. A locally bounded weak solution with initial data g must satisfy

$$\int_0^\infty \int_{\mathbb{R}} (u\varphi_t + F(u)\varphi_x)\,dx\,dt + \int_{\mathbb{R}} g(x)\varphi(x,0)\,dx = 0$$

for every smooth test function compactly supported in $\mathbb{R} \times [0,\infty)$. The initial term follows from time integration by parts; compact support removes the spatial boundary term.

For a piecewise constant state u_L to the left of $x = st$ and u_R to the right, the delta coefficient in $u_t + F(u)_x$ is $-s(u_R - u_L) + F(u_R) - F(u_L)$. It vanishes precisely when

$$s(u_R - u_L) = F(u_R) - F(u_L).$$

This is the Rankine–Hugoniot condition. For Burgers with $u_L \neq u_R$, it gives $s = (u_L + u_R)/2$. Yet it permits both compressive and expansive

爆破的量必须说明

Burgers 特征线上 $u = g(\xi)$ 始终有界，失控的是 u_x。这叫梯度爆破，不能误写成 $|u| \to \infty$。局部存在定理的延拓判据通常取决于某个特定范数，而不是笼统的“解变大”。

jumps. The entropy criterion selects a compressive shock, $u_L > u_R$, for which $u_R < s < u_L$; a jump with $u_L < u_R$ is not the entropy solution. This calculation explains why merely imposing the weak identity is insufficient. The general entropy theory belongs to the later chapters.

1.4 Overview

This textbook is divided into three major Parts.

Part I: Representation Formulas for Solutions

Here we identify those important partial differential equations for which in certain circumstances explicit or more-or-less explicit formulas can be had for solutions. The general progression of the exposition is from direct formulas for certain linear equations to far less concrete representation formulas, of a sort, for various nonlinear PDE.

Chapter 2 is a detailed study of four exactly solvable partial differential equations: the linear transport equation, Laplace's equation, the heat equation, and the wave equation. These PDE, which serve as archetypes for the more complicated equations introduced later, admit directly computable solutions, at least in the case that there is no domain whose boundary geometry complicates matters. The explicit formulas are augmented by various indirect, but easy and attractive, "energy"-type arguments, which serve as motivation for the developments in Chapters 6, 7 and thereafter.

Chapter 3 continues the theme of searching for explicit formulas, now for general first-order nonlinear PDE. The key insight is that such PDE can, locally at least, be transformed into systems of ordinary differential equations (ODE), the characteristic equations. We stipulate that once the problem becomes "only" the question of integrating a system of ODE, it is in principle solved, sometimes quite explicitly. The derivation of the characteristic equations given in the text is very simple and does not require any geometric insights. It is in truth so easy to derive the characteristic equations that no real purpose is had by dealing with the quasilinear case first.

We introduce also the Hopf–Lax formula for Hamilton–Jacobi equations (§3.3) and the Lax–Oleinik formula for scalar conservation laws (§3.4). (Some knowledge of measure theory is useful here but is not essential.) These sections provide an early acquaintance with the global theory of these important nonlinear PDE and so motivate the later Chapters 10 and 11.

Chapter 4 is a grab bag of techniques for explicitly (or kind of explicitly) solving various linear and nonlinear partial differential equations, and the reader should study only whatever seems interesting. The section on the Fourier transform is, however, essential. The Cauchy–Kovalevskaya Theorem appears at the very end. Although this is basically the only general existence theorem in the subject, and thus logically should perhaps be regarded as central, in practice these power series methods are not so prevalent.

局部存在不等于稳定

Cauchy–Kovalevskaya 定理要求解析系数、解析初值及非特征初始面，给出局部解析解。它不直接给出光滑或 Sobolev 数据下的连续依赖，也不保证全局解。

Part II: Theory for Linear Partial Differential Equations

Next we abandon the search for explicit formulas and instead rely on functional analysis and relatively easy "energy" estimates to prove the existence of weak solutions to various linear PDE. We investigate also the uniqueness and regularity of such solutions and deduce various other properties.

Chapter 5 is an introduction to Sobolev spaces, the proper setting for the study of many linear and nonlinear partial differential equations via energy methods. This is a hard chapter, the real worth of which is only later revealed, and requires some basic knowledge of Lebesgue measure theory. However, the requirements are not really so great, and the review in Appendix E should suffice. In my opinion there is no particular advantage in considering only the Sobolev spaces with exponent $p = 2$, and indeed insisting upon this obscures the two central inequalities, those of Gagliardo–Nirenberg–Sobolev (§5.6.1) and of Morrey (§5.6.2).

指数与维数

Sobolev 空间的指数不能随意固定。当 $1 \le p < n$，临界指数 $p^* = np/(n-p)$ 控制可积性增益；当 $p > n$，Morrey 估计给出 Hölder 连续性。维数在这里直接决定可得的正则性。

In Chapter 6 we vastly generalize our knowledge of Laplace's equation to other second-order elliptic equations. Here we work through a rather complete treatment of existence, uniqueness and regularity theory for solutions, including the maximum principle, and also a reasonable introduction to the study of eigenvalues, including a discussion of the principal eigenvalue for nonselfadjoint operators.

Chapter 7 expands the energy methods to a variety of linear partial differential equations characterizing evolutions in time. We broaden our earlier investigation of the heat equation to general second-order parabolic PDE and of the wave equation to general second-order hyperbolic PDE. We study as well linear first-order hyperbolic systems, with the aim of motivating the developments concerning nonlinear systems of conservation laws in Chapter 11. The concluding Section 7.4 presents the alternative functional analytic method of semigroups for building solutions.

(Missing from this long Part II on linear partial differential equations is any discussion of distribution theory or potential theory. These are

important topics, but for our purposes seem dispensable, even in a book of such length. These omissions do not slow us up much and make room for more nonlinear theory.)

Part III: Theory for Nonlinear Partial Differential Equations

This section parallels for nonlinear PDE the development in Part II but is far less unified in its approach, as the various types of nonlinearity must be treated in quite different ways.

Chapter 8 commences the general study of nonlinear partial differential equations with an extensive discussion of the calculus of variations. Here we set forth a careful derivation of the direct method for deducing the existence of minimizers and discuss also a variety of variational systems and constrained problems, as well as minimax methods. Variational theory is the most useful and accessible of the methods for nonlinear PDE, and so this chapter is fundamental.

Chapter 9 is, rather like Chapter 4 earlier, a gathering of assorted other techniques of use for nonlinear elliptic and parabolic partial differential equations. We encounter here monotonicity and fixed point methods and a variety of other devices, mostly involving the maximum principle. We study as well certain nice aspects of nonlinear semigroup theory, to complement the linear semigroup theory from Chapter 7.

不同的广义解机制

守恒律用积分恒等式表达守恒，再加熵不等式；Hamilton–Jacobi 方程的黏性解用接触试验函数表达微分不等式。两者的稳定性和唯一性机制不同，不能仅以“弱导数存在”统一定义。

Chapter 10 is an introduction to the modern theory of Hamilton–Jacobi PDE and in particular to the notion of “viscosity solutions”. We encounter also the connections with the optimal control of ODE, through dynamic programming.

Chapter 11 picks up from Chapter 3 the discussion of conservation laws, now systems of conservation laws. Unlike the general theoretical developments in Chapters 5–9, for which Sobolev spaces provide the proper abstract framework, we are forced to employ here direct linear algebra and calculus computations. We pay particular attention to the solution of Riemann’s problem and to entropy criteria.

Chapter 12, an introduction to nonlinear wave equations, is new with this edition. We provide long time and short time existence theorems for certain quasilinear wave equations and an in-depth examination of semilinear wave equations, especially for subcritical and critical power nonlinearities in three space dimensions. To complement these existence theorems, the final section identifies various criteria ensuring nonexistence of solutions.

Appendices A–E provide for the reader’s convenience some background material, with selected proofs, on inequalities, linear functional

analysis, measure theory, etc.

The Bibliography is an updated and extensive listing of interesting PDE books to consult for further information. Since this is a textbook and not a reference monograph, I have mostly not attempted to track down and document the original sources for the myriads of ideas and methods we will encounter. The mathematical literature for partial differential equations is truly vast, but the books cited in the Bibliography should at least provide a starting point for locating the primary sources. (Citations to selected research papers appear throughout the text.)

1.5 Problems

Source problems (Evans, Chapter 1, Problems 1–5)

The numbering in this list is the source problem numbering. Solutions in the next subsection are editorial additions.

1. Classify each of the partial differential equations in §1.2 as follows:
   (a) Is the PDE linear, semilinear, quasilinear or fully nonlinear?
   (b) What is the order of the PDE?
2. Let k be a positive integer. Show that a smooth function defined on $\mathbb{R}^n$ has in general
$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}$$
distinct partial derivatives of order k.
(Hint: This is the number of ways of inserting $n-1$ dividers | within a row of k symbols $\circ$: for example, $\circ\circ \,||\, \circ\circ\circ \,|\, \circ \,|\, \circ\circ\circ \,||\, \circ\circ\circ\circ|$. Explain why each such pattern corresponds to precisely one of the partial derivatives of order k.)

The next exercises provide some practice with the multiindex notation introduced in Appendix A.

3. Prove the Multinomial Theorem:
$$(x_1 + \cdots + x_n)^k = \sum_{|\alpha|=k} \binom{|\alpha|}{\alpha} x^\alpha,$$
where $\binom{|\alpha|}{\alpha} := |\alpha|!/\alpha!$, $\alpha! = \alpha_1! \cdots \alpha_n!$, and $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$. The sum is taken over all multiindices $\alpha = (\alpha_1, \ldots, \alpha_n)$ with $|\alpha| = k$.
4. Prove Leibniz's formula:
$$D^\alpha(uv) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} D^\beta u D^{\alpha-\beta} v,$$

where $u, v : \mathbb{R}^n \to \mathbb{R}$ are smooth, $\binom{\alpha}{\beta} := \alpha!/[\beta!(\alpha-\beta)!]$, and $\beta \le \alpha$ means $\beta_i \le \alpha_i$ $(i = 1, \dots, n)$.

5. Assume that $f : \mathbb{R}^n \to \mathbb{R}$ is smooth. Prove

$$f(x) = \sum_{|\alpha| \le k} \frac{1}{\alpha!} D^\alpha f(0) x^\alpha + O(|x|^{k+1}) \qquad \text{as } x \to 0$$

for each $k = 1, 2, \dots$. This is Taylor's formula in multiindex notation. (Hint: Fix $x \in \mathbb{R}^n$ and consider the function of one variable $g(t) := f(tx)$.)

Editorial supplement: solutions to Problems 1–5

Problem 1: classification. The following tables use the most restrictive applicable class, for generic nonlinearities. The order is the largest total derivative order in space and time. Coefficients and forcing terms are prescribed, not additional unknowns, unless explicitly stated otherwise.

Scalar linear equation (§1.2.1a)	Class	Order
1. Laplace	Linear	2
2. Helmholtz	Linear	2
3. Transport	Linear	1
4. Liouville	Linear	1
5. Heat	Linear	2
6. Schrödinger	Linear	2
7. Kolmogorov	Linear	2
8. Fokker–Planck	Linear	2
9. Wave	Linear	2
10. Klein–Gordon	Linear	2
11. Telegraph	Linear	2
12. General wave	Linear	2
13. Airy	Linear	3
14. Beam	Linear	4

Scalar nonlinear equation (§1.2.1b)	Class	Order
1. Eikonal	Fully nonlinear	1
2. Nonlinear Poisson	Semilinear	2
3. p-Laplacian	Quasilinear	2
4. Minimal surface	Quasilinear	2
5. Monge–Ampère	Fully nonlinear	2
6. Hamilton–Jacobi	Fully nonlinear	1
7. Scalar conservation law	Quasilinear	1
8. Inviscid Burgers	Quasilinear	1
9. Reaction-diffusion	Semilinear	2
10. Porous medium	Quasilinear	2
11. Nonlinear wave	Semilinear	2
12. KdV	Semilinear	3
13. Nonlinear Schrödinger	Semilinear	2

The p-Laplacian entry assumes $1 < p < \infty$, $p \neq 2$, using its classical expansion away from $Du = 0$ as in §1.2. The porous medium entry assumes $\gamma > 1$ and uses the expansion where $u > 0$. The Monge–Ampère entry assumes $n \geq 2$ and given f. If $H(q, x)$ is affine in q, Hamilton–Jacobi is quasilinear (in fact linear in this particular displayed form). If $f(u)$ is affine, the nonlinear Poisson, reaction-diffusion and nonlinear wave equations become linear. For the Schrödinger equations the unknown is complex-valued; one may equivalently use its real and imaginary parts as a real system.

先约定参数与分支

表中按一般非线性列出。$p = 2$ 时 p-Laplacian 变为线性；$\gamma = 1$ 时多孔介质式变为热方程；$n = 1$ 且 f 已知时 Monge–Ampère 式是 $u_{xx} = f$。忽略这些例外会把分类误当成方程名称的固定属性。

System (§1.2.2)	Class	Order
a1. Elastic equilibrium	Linear	2
a2. Elastic evolution	Linear	2
a3. Maxwell	Linear	1
b1. Conservation laws	Quasilinear	1
b2. Reaction-diffusion	Semilinear	2
b3. Incompressible Euler	Quasilinear	1
b4. Incompressible Navier–Stokes	Semilinear	2

For a smooth scalar flux, the chain rule gives

$$\operatorname{div} \mathbf{F}(u) = \sum_i (F^i)'(u) u_{x_i},$$

hence its quasilinear form. For a system with m components, $\mathbf{F} : \mathbb{R}^m \to \mathbb{R}^{m \times n}$ and $(\operatorname{div} \mathbf{F}(\mathbf{u}))^a = \sum_i \partial_{x_i} F^{ai}(\mathbf{u})$. Affine fluxes give linear equations or systems.

For the incompressible flow entries, the unknowns are $(\mathbf{u}, p)$. Navier–Stokes has constant coefficients on the second derivatives of $\mathbf{u}$; its nonlinear convection term has order one. This is the sense of the semilinear classification in the table. The divergence equation has order one, and there

are no second derivatives of p in the displayed system. It is a system of mixed differential orders, not an unconstrained uniformly parabolic system for all $n+1$ unknowns. Euler has first-order derivatives with coefficients depending on $\mathbf{u}$, hence is quasilinear.

Problem 2: counting independent derivative operators.

Expanded proof (editorial solution to source Problem 2).

For a C^k function, the order of differentiation does not matter. Thus a distinct derivative operator of order k is determined by a multiindex $\alpha \in \{0,1,2,\ldots\}^n$ satisfying $\alpha_1 + \cdots + \alpha_n = k$. Represent α_i by α_i circles in the ith compartment between $n-1$ dividers. Empty compartments are allowed: they encode $\alpha_i = 0$. This is a bijection.

Among the $k+n-1$ positions, choose the $n-1$ divider positions, giving $\binom{k+n-1}{n-1} = \binom{k+n-1}{k}$. For $n = 1$ there are no dividers and exactly one derivative, as the same formula predicts. This counts operators; special functions may have several derivatives with equal values. □

Problem 3: the multinomial theorem.

Expanded proof (editorial solution to source Problem 3).

Expand the product of k copies of $(x_1 + \cdots + x_n)$ by distributivity. A term x^α is obtained by selecting x_i from exactly α_i of the factors. First choose α_1 of the k factors, then α_2 of the remaining factors, and so on. The number of such choices is

$$\binom{k}{\alpha_1}\binom{k-\alpha_1}{\alpha_2}\cdots\binom{\alpha_n}{\alpha_n} = \frac{k!}{\alpha_1!\cdots\alpha_n!}.$$

Grouping all terms with the same multiindex proves the formula. The identity also holds for $k = 0$, with the single multiindex $\alpha = 0$. □

两类组合系数

多项式展开中 $\binom{k}{\alpha} = k!/\alpha!$，要求 $|\alpha| = k$。乘积求导中 $\binom{\alpha}{\beta} = \prod_i \binom{\alpha_i}{\beta_i}$，逐坐标分配导数。后者一般不等于 $\binom{|\alpha|}{|\beta|}$。

Problem 4: the multiindex product rule.

Expanded proof (editorial solution to source Problem 4).

Apply the one-variable product rule α_i times in coordinate x_i, for each $i = 1,\ldots,n$. If exactly β_i of these derivatives fall on u, the remaining $\alpha_i - \beta_i$ fall on v, and the coefficient is $\binom{\alpha_i}{\beta_i}$. Since mixed derivatives commute under the smoothness assumption, the resulting term is $D^\beta u\, D^{\alpha-\beta} v$, with coefficient

$$\prod_{i=1}^{n}\binom{\alpha_i}{\beta_i} = \frac{\alpha!}{\beta!(\alpha-\beta)!}.$$

Summing over all coordinatewise choices $0 \le \beta_i \le \alpha_i$ proves the identity. For $\alpha = 0$ it reduces to $uv = uv$. □

Problem 5: Taylor's formula with a uniform remainder.

Expanded proof (editorial solution to source Problem 5).

It suffices that $f \in C^{k+1}$ on a neighborhood of 0. Choose $r > 0$ so the closed ball $\overline{B}_r(0)$ lies inside this neighborhood. For $|x| < r$ set $g_x(t) = f(tx)$, $0 \le t \le 1$. Repeated differentiation and the multinomial theorem give, for $0 \le j \le k+1$,

$$g_x^{(j)}(t) = \left(\sum_{i=1}^{n} x_i \partial_{x_i}\right)^j f(tx) = \sum_{|\alpha|=j} \frac{j!}{\alpha!} x^\alpha D^\alpha f(tx).$$

The one-variable Taylor formula with integral remainder gives

$$f(x) = \sum_{j=0}^{k} \frac{g_x^{(j)}(0)}{j!} + R_k(x),$$
$$R_k(x) = \frac{1}{k!} \int_0^1 (1-t)^k g_x^{(k+1)}(t)\, dt.$$

Substituting the derivative formula converts the finite sum into

$$\sum_{|\alpha|\le k} \frac{D^\alpha f(0)}{\alpha!} x^\alpha.$$

Continuity and compactness give a finite common bound

$$M = \max_{\substack{|\alpha|=k+1 \\ y\in\overline{B}_r(0)}} |D^\alpha f(y)|.$$

Using $|x^\alpha| \le |x|^{k+1}$ and $\int_0^1 (1-t)^k\, dt = 1/(k+1)$, we obtain

$$|R_k(x)| \le M \left(\sum_{|\alpha|=k+1} \frac{1}{\alpha!}\right) |x|^{k+1}.$$

The constant depends only on n, k and the indicated local derivative bound, not on the direction of x. This proves the required $O(|x|^{k+1})$. □

1.6 References

Klainerman's article [Kl] is a nice modern overview of the field of partial differential equations.

Good general texts and monographs on PDE include Arnold [Ar2], Courant–Hilbert [C-H], Craig [Cr], DiBenedetto [DB1], Folland [F1], Friedman [Fr2], Garabedian [G], John [J2], Jost [Jo], Levy–Shearer [L-S] McOwen [MO], Mikhailov [M], Olver [O2] Petrovsky [Py], Rauch [R1], Renardy–Rogers [R-R], Showalter [Sh], Smirnov [Sm], Smoller [S], Strauss [St2], Taylor [Ta], Thoe–Zachmanoglou [T-Z], Vasy [Vy], Zauderer [Za], and many others. The prefaces to Arnold [Ar2] and to Bernstein [Bt] are particularly interesting reading. Zwillinger's handbook [Zw] on differential equations is a useful compendium of methods for PDE.

Editorial source note. The bracketed keys above are Evans's bibliography identifiers, retained as source identifiers rather than local citation numbers. Their full entries are in the supplied English source, printed p. 689 onward (PDF page 706 onward). This list records the supplied printing; it is not an updated literature survey. The supplied Chinese printing has a shorter list, omitting, for example, Craig [Cr], Jost [Jo], Levy–Shearer [L-S], Olver [O2], and Vasy [Vy]. The English printing controls the transcription here.

余项须对方向一致

逐条射线使用一元 Taylor 公式还不够：余项常数可能依赖方向。在固定闭球上统一控制所有 $k+1$ 阶导数，才得到真正的多元 $O(|x|^{k+1})$。仅假设 C^k 通常只能得到 $o(|x|^k)$。